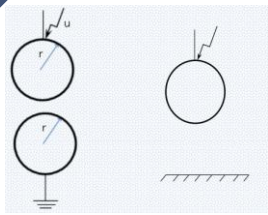
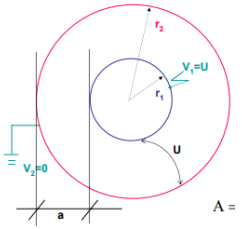
EE4047 – High Voltage Techniques I

New Topic:
Spherical Electrode Systems





Week 7.1 – Monday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

10.04.2023

OUTLINE

1 Concentric Spherical Electrode Systems

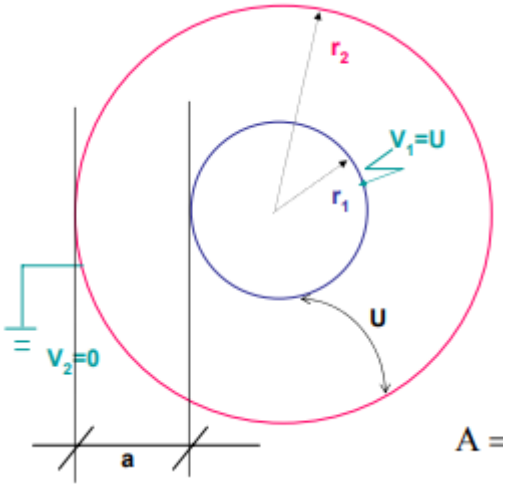
2 Multilayer Dielectric Concentric Spheres

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

An electrode system that consists of two metalic spheres that has the same center, with a dielectric material between them.

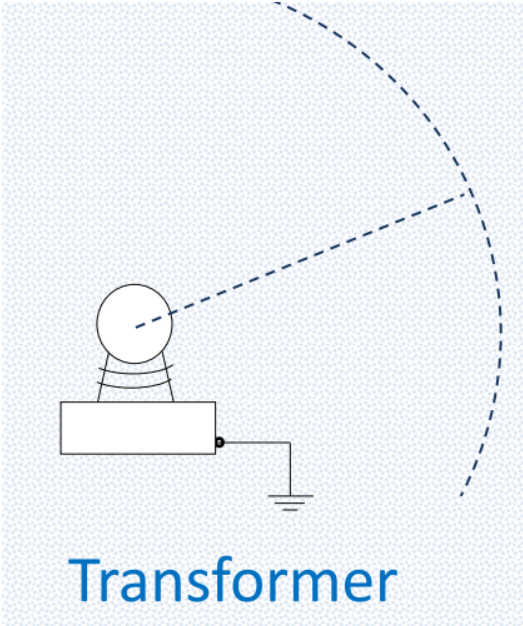
Although it does not have direct applications in the field, the shape and positioning of many electrodes require design and analysis assuming a concentric spherical electrode systems.

Spherical electrodes or electrodes which have shape similar to a sphere are widely used for reducing electric field strength in an area or take precautions for electric discharge events.



1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

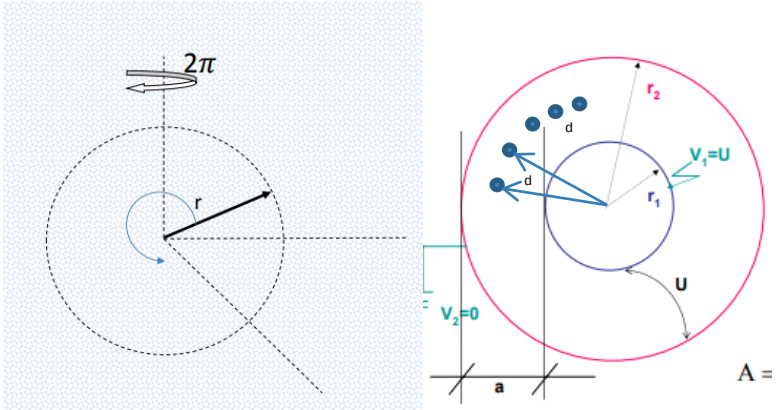
Considering such systems as a concentric spherical electrode system which has an outer electrode with a very large radius simplifies the solutions.



1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Due to rotational symmetry of sphere, potential V which changes in 3 dimensions can be analyzed in one dimension with respect to radius r, the distance from the center of the system.

$$\frac{d^2V}{dr^2}+\frac{2}{r}\frac{dV}{dr}=0 \qquad V=A+\frac{B}{r}$$



Solving A and B constants using the boundary conditions as $V = V_1 = U$ for $r = r_1$, and $V = V_2 = 0$ for $r = r_2$

$$V = U\frac{r_1}{r_2 - r_1}\left(\frac{r_2}{r} - 1\right)$$

Negative gradient of which gives us electric field strength:

$$E = U\frac{r_1.r_2}{r_2 - r_1}.\frac{1}{r^2}$$

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Due to symmetricity of the spherical electrode system, the flux density leaving the surface of a spherical electrode with r radius can be calculated by replacing the S in the equation

$$D = \frac{Q}{S}$$

with the equation that gives the surface of a sphere.

$$D = \frac{Q}{4\pi r^2}$$

Electric field strength can be derived using the relation $D = \varepsilon E$

$$E = \frac{Q}{4\pi r^2}\frac{1}{r^2}$$

Integral of elecric field strength equation gives us the voltage applied across the two concentric spherical electrodes:

$$U = \frac{Q}{4\pi\varepsilon}\frac{r_2 - r_1}{r_1.r_2} \quad \leftrightarrow \quad \frac{Q}{4\pi\varepsilon} = U\frac{r_1.r_2}{r_2 - r_1}$$

CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

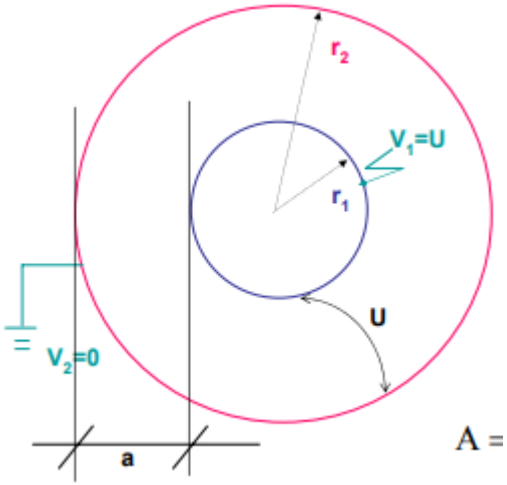
Using the equation for electric field strength, the maximum, minimum and average electric field strengths can be found.

The general electric field strength formula we derived in the previous slides:

$$E = U\frac{r_1.r_2}{r_2 - r_1}.\frac{1}{r^2} \quad \rightarrow \quad E = E_{max} = \frac{U}{\frac{r_1}{r_2}(r_2 - r_1)}$$

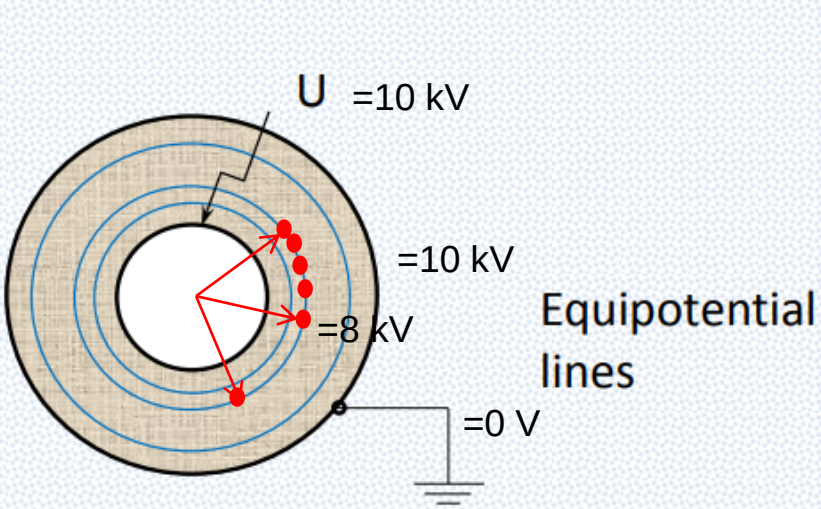
$$r = r_2 \quad \rightarrow \quad E = E_{min} = \frac{U}{\frac{r_2}{r_1}(r_2 - r_1)}$$

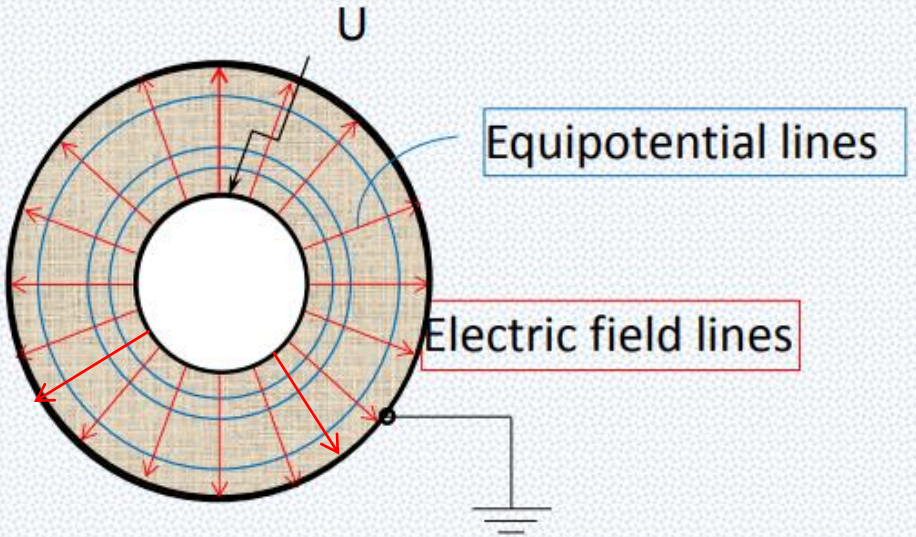
$$r = r_{ave} = \sqrt{r_1.r_2} \quad \rightarrow \quad E = E_{ave} = U\frac{r_1.r_2}{r_2 - r_1}.\frac{1}{{r_{ave}}^2} = \frac{U}{r_2 - r_1}$$



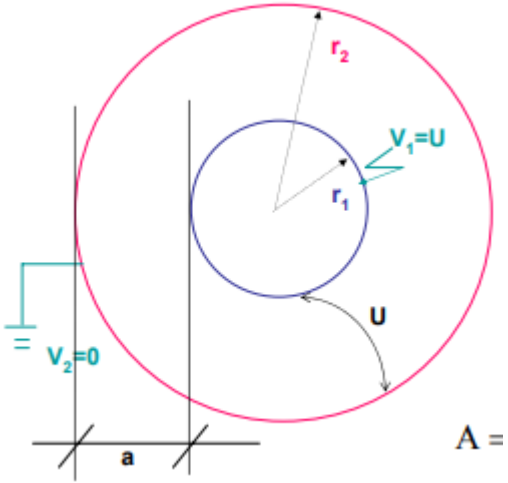
1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Electric field lines ALWAYS perpendicular to equipotential lines (ORTHOGONAL).

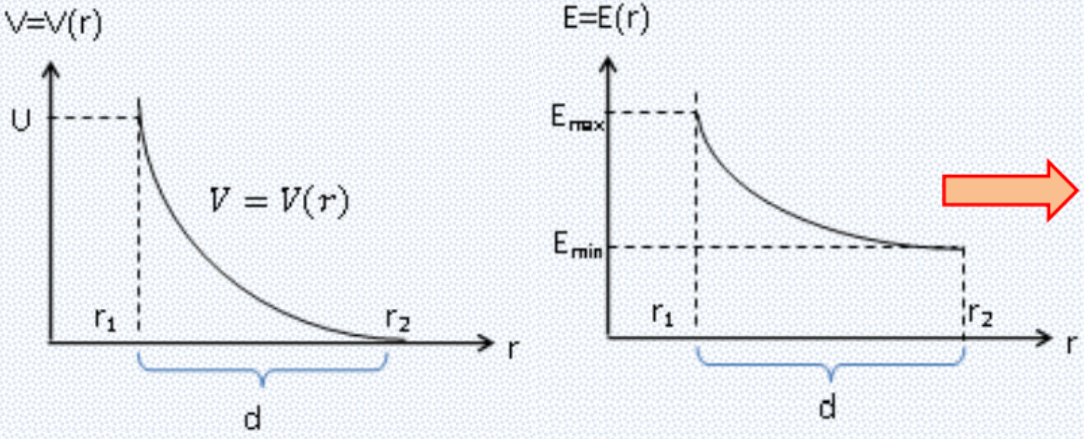




1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS



A =



Does it look uniform to you?

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

Capacity of a concentric spherical electrode system can be calculated using the voltage equation we previously derived using the sphere surface area and charge. The unit for the inner shpere radius r_1 and out sphere radius r_2 should be (m), dielectric constant unit should be (F/m). $\varepsilon_0 = 8.854.10^{-12}\ F/m$ constant can be used and the result will be in Farad.

$$U = \frac{Q}{4\pi\varepsilon}\frac{r_2 - r_1}{r_1.r_2}$$

$$C = \frac{Q}{U} = 4\pi\varepsilon\frac{r_1.r_2}{r_2 - r_1} = 4\pi\varepsilon_r\varepsilon_0\frac{r_1.r_2}{r_2 - r_1}$$

1 CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

When defining, calculating and comparing different spherical electrode systems, we use a number of parameters.

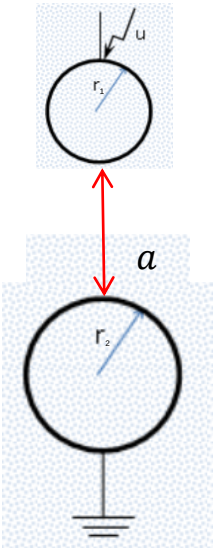
Actual electrode seperation, $a = r_2 - r_1$

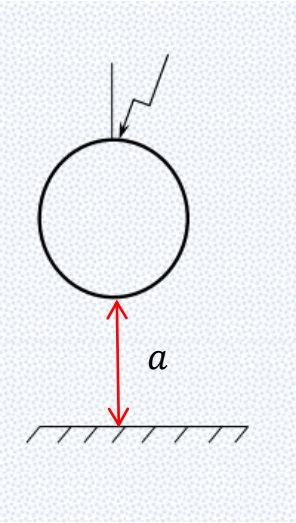
Effective electrode seperation, $\alpha = \frac{U}{E_{max}} = \frac{r_1}{r_2}(r_2 - r_1)$

Geometric characteristics, $p = \frac{r_1 + a}{r_1} = \frac{r_2}{r_1}$ $\quad q = \frac{r_2}{r_1} = p$

For concentric SES

Utilization factor (electrode efficiency), $\eta = \frac{\alpha}{a} = \frac{\frac{r_1}{r_2}(r_2 - r_1)}{(r_2 - r_1)} = \frac{r_1}{r_2} = \frac{1}{p}$





CONCENTRIC SPHERICAL ELECTRODE SYSTEMS

If utilization factor (η), actual electrode seperation (a) and the voltage applied to the system are known, E_{max} can be calculated through

$U = E_{max}.\alpha = E_{max}.a.\eta$

If $E_{max} > E_{breakdown}$ the discharge starts closer to the spherical electrode with the smaller radius.

In nonuniform electric fields, breakdown does not happen immediately. Primarily partial discharges take place. As voltage increases these discharges become more frequent and after passing a number of stages full breakdown is encountered. This process depends on the level of nonuniformity of the field.

Level of nonuniformity (inverse of utilization factor): $\frac{1}{\eta}$ (in concentric SES, equal to p)

END OF THE LESSON

Any questions?

Check out our playlist s on Youtube:

High Voltage Technique: https://www.youtube.com/playlist?list=PLLZI7WaEmZi04KMWVURGlU34W3LoNZyWX

Insulators: https://www.youtube.com/playlist?list=PLLZI7WaEmZi1ReLGq0qfP1VjRqxDJi5kA

Lightning Impulse Voltage: https://www.youtube.com/playlist?list=PLLZI7WaEmZi2isWOB-zBbb6LNplH28byQ